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Peer review should constrain evaluative authority

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Without reliable ground truth for a “good” or “right” review, peer-review reform should constrain how uncertain judgments gain consequential authority.

Why is peer review under pressure?

Peer review determines which submissions enter the scientific record.

Academic opportunities



Publication shapes careers, funding, recognition, and academic qualifications.

Unmatched scale



Submissions are growing faster than the pool of qualified reviewers.

Weak reviewing incentives



Publishing advances careers. Careful reviewing earns much less recognition.

Competing interests



Researchers judge papers while competing for publication, funding, and career opportunities.

What can’t we fix?

At decision time, no definitive answer key for a “good” or “right” review.

Different standards



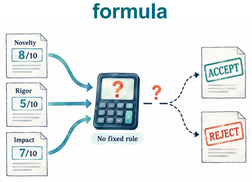
Disagreement can persist even when both reviewers act in good faith.

Opaque motives



We see the review. We may not see the motive.

No fixed decision formula



No fixed formula maps criteria to recommendations or final decisions.

Why aren’t existing reforms enough?

No ground truth. No guarantee of a “good” or “right” review.



Many-signal review systems

More reviews do not ensure timely correction.

Early judgments can anchor decisions before rebuttal. More reviews do not eliminate disagreement or stop one negative review from outweighing positive ones.



Human-centered improvement

Training cannot eliminate subjectivity or ensure compliance.

Standards still differ, incentives can override rules, and reviewer expertise cannot grow as fast as submissions.



AI-assisted review

A capable AI still lacks a reliable learning target.

Past reviews have no verified ground truth; inconsistent, low-effort or strategic reviews contaminate training.

The key problem is not only the review quality.
It is unchecked evaluative authority under persistent uncertainty.

What do we propose to change?

Consequential judgments must face scrutiny before influencing decisions.



01 Separate critique from authority
Writing a critique does not grant unilateral power to turn it into a consequential score.

02 Require an independent check
Inspect support and interpretation before consequential influence.

03 Allow challenge and correction
Clarify, narrow, downgrade or reverse unsupported judgments.

04 Keep an accountable record
Preserve consequential claims and corrections in the record.

One concrete instantiation

Human–AI mutual supervision

Neither humans nor AI may turn their own critique into a consequential score.

01 CRITIQUE

Track A — AI → Humans

AI writes critique
No consequential score

Track B — Humans → AI

Humans write critiques
No consequential score

02 INDEPENDENT SCRUTINY

Humans check, then judge
Record agreement, disagreement and important omissions.

AI checks and infers a rating
Check evidence, interpretation, and proportionality.

03 FINAL ADJUDICATION



Chair / editor
Adjudicate separate outputs. No automatic averaging.

Unresolved claims: clarify, reduce influence or withhold it.

Independent checks and possible correction FIRST → **THEN decision influence**

Initial one-case feasibility signal

One paper · four human supervisors · four existing reviews · one AI · one chair.

Usable counter-signals	4 / 4 found it easier	3 / 4 perceived gains	Chair workload uncertain
AI: 3→6, 6→8, 6→6, 8→8. Human-after-AI: 5, 5, 5, 6. Flags: superficiality, inaccuracy, misguidance.	Two consulted the manuscript; two did not. Supervision felt easier than full reviewing.	Three perceived better output; one worse. Chair reported no change in output quality.	More signals to assess, but unsupported or conflicting judgments may be easier to identify. Final decision: Reject.

One case only; decision quality and benefits at scale still need validation.

How far are we from the goal?

Start with AI scrutiny, then test the full dual-track design.

1. Resolve technical challenges
2. Connect research teams
3. Propose workshops
4. Partner with a real venue

Technical collaboration: scrutiny, datasets, AI supervision, HCI and security.

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